# seqhesion: a label-blind hierarchy of fungal ITS sequences from many small overlapping trees

Draft

Josh Walker
Independent Researcher

Claude Code
(Claude Opus 5.5, Anthropic)

Draft of 18 October 2026

**Abstract**

Community DNA-barcoding programmes now sequence the nuclear ribosomal internal transcribed spacer (ITS) of tens of thousands of vouchered fungal collections, often far faster than taxonomists can name them. Many of these sequences carry provisional, inconsistent or missing names, so any grouping that leans on the names inherits their errors. seqhesion ("sequence cohesion") groups sequences from sequence evidence alone. It builds thousands of small phylogenetic trees (*shards*) over overlapping neighbourhoods of about 150 sequences each. For every pair of sequences, each shard that holds both records the diameter of the smallest clade containing them; the median over shards is the pair's join level. Average linkage over the observed pairs only gives a hierarchy cut at fixed tree-distance levels. Every group carries its evidence: cohesion, pull towards outside groups, and their difference, the *margin*, which predicted which groups recur when the shard trees are resampled (AUC 0.94). Coarser structure comes from separate layers of stratified sparse shards, first within each connected component and then across components. Group identities (*antenomina*) persist across releases by a mutual-best-match rule, and partial ITS2-only sequences are placed onto the stored shard alignments without changing the hierarchy. On a corpus of 225,985 input sequences, drawn mainly from MycoMap with other sources (146,946 distinct full-ITS sequences in 4,174 components), seqhesion produced 51,269 fine-level groups with per-group confidence. It is an evidence layer and a navigation aid for curators, not a phylogeny. This paper describes the method, the choices behind it and their limits, and the in-house experiments that support them.

## 1 Introduction

The ITS region is the formal DNA barcode for Fungi (Schoch et al., 2012), and community science has made it cheap to sequence. Programmes such as MycoMap sequence vouchered macrofungal collections made by volunteers, increasingly with nanopore long reads, and publish the consensus sequences with whatever name the collector or an identification tool assigned. The result is a large, fast-growing corpus in which much of the diversity is undescribed or only provisionally named, many names are field identifications, and the same taxon may appear under several names, or several taxa under one.

Two common ways of organising such a corpus both depend on names or fixed identity thresholds. Reference-based identification assigns each sequence the name of its closest match. Threshold clustering, such as the species hypotheses of UNITE (Kõljalg et al., 2013; Abarenkov et al., 2024), groups sequences by single-linkage identity at a few fixed distances. Studies of ITS variation show that no single identity threshold separates species across the fungal kingdom (Nilsson et al., 2008; Vu et al., 2019). Even within one genus such as *Cortinarius*, the best threshold varies by lineage (Garnica et al., 2016).

seqhesion takes a different position. It is an *evidence layer*: it groups sequences purely from the sequence data and reports, for every group, how strongly the data support it. Names never enter any decision. Labels may travel with the input for display, but nothing in the processing path

reads them. Choosing which groups become named units is curation, and happens downstream in a separate tool that maps names onto seqhesion's groups. Agreement with names is used only as a sanity check, never as a target to optimise. We call the design *label-blind*.

Being label-blind has three practical consequences.

1. The groups cannot inherit misidentifications. A curator who finds a name scattered over several well-supported groups learns something about the name.
2. The output does not depend on the state of the taxonomy. Undescribed lineages are first-class groups with their own persistent identifiers (*antenomina*, "before the name", Section 3.9), to which comments and curation can attach before any name exists.
3. Validation must use something other than names. We use reproducibility under independent resampling of the shards, and treat name agreement only as a display.

The second idea is to build from many small trees rather than one large one. A single alignment of tens of thousands of ITS sequences is not credible: ITS is alignable only among fairly close relatives. A single tree, even if feasible, gives one topology with no indication of which parts would survive a rebuild. seqhesion instead builds thousands of small trees over overlapping neighbourhoods, each from a carefully aligned set of about 150 related sequences. It then pools what the trees say about every pair of sequences. The overlap is the source of the confidence measures. Pairs that no shard holds are treated as missing evidence, not as evidence of distance.

This paper is a first description of the method and its current results. Section 2 places it in the literature. Section 3 describes the pipeline, and Section 4 its implementation and reproducibility guarantees. Section 5 reports in-house experiments, and Section 6 states the limitations plainly.

## 2 Related work

**ITS barcoding and reference clustering.** The ITS region was proposed as the primary fungal barcode for its high rate of successful identification across the kingdom (Schoch et al., 2012), building on the DNA barcoding programme of Hebert et al. (2003). UNITE organises ITS sequences into species hypotheses (SHs) by clustering at a series of distance thresholds within higher-level groups, with experts choosing the threshold for each lineage (Kõljalg et al., 2013; Abarenkov et al., 2024). The Barcode Index Number system of BOLD gives animal COI clusters persistent registry identifiers (Ratnasingham and Hebert, 2013). Our antenomina serve a similar purpose for tree-based groups. Studies of intraspecific ITS variability show that fixed identity thresholds misclassify many species (Nilsson et al., 2008; Vu et al., 2019; Garnica et al., 2016). Divergent intragenomic ITS copies are common (Bradshaw et al., 2023), and apparent barcode gaps shrink as sampling becomes more complete (Meyer and Paulay, 2005). seqhesion uses identity only to decide which sequences are sampled together, never to group them, and reports structure at several levels instead of choosing one threshold.

**Single-locus species delimitation.** Distance-based methods such as ABGD (Puillandre et al., 2012) and ASAP (Puillandre et al., 2021) find gaps in pairwise distance distributions. ASAP returns a ranked list of partitions from one hierarchical clustering, the closest analogue to our multi-level output. Tree-based methods (GMYC (Pons et al., 2006; Fujisawa and Barraclough, 2013), PTP and bPTP (Zhang et al., 2013), mPTP (Kapli et al., 2017)) fit models of speciation and coalescence to a single tree. They need one tree over all sequences, and some need it binary or ultrametric. They return a single partition, possibly with per-node support. seqhesion does not attempt species delimitation. It reports a nested set of groups, each with its evidence, and leaves the choice of level, clade by clade, to curators.

**Divide-and-conquer phylogenetics and supertrees.** Large-scale alignment and tree estimation are commonly divided into subproblems: DACTAL (Nelesen et al., 2012) and PASTA (Mirarab et al., 2015) work on overlapping or disjoint subsets and merge them. uDance (Balaban et al., 2024), used for Greengenes2 (McDonald et al., 2024), grows trees by placement and refines them per partition.

Supertree methods combine trees on overlapping taxon sets. These include Min-Cut (Semple and Steel, 2000), the modified Min-Cut of Page (2002), which treats a polytomy as missing information rather than conflict, the Greedy Strict Consensus Merger (Fleischauer and Böcker, 2016) and the Spectral Cluster Supertree (McArthur et al., 2024). The last links taxa only when some input tree holds both, so a pair never seen together is no evidence. seqhesion shares that principle. It differs from these methods in its goal: it does not try to produce a single resolved tree, only a hierarchy of groups with their evidence. It also never merges trees: it pools what each tree says about pairs.

**Distance-based supertrees and fit to the input trees.** The closest relatives of seqhesion are the distance-based supertree methods. Average Consensus (Lapointe and Cucumel, 1997) turns each input tree into path-length distances between its taxa, averages the matrices, and fits one tree to the average. SDM (Criscuolo et al., 2006) first rescales the input matrices to a common scale, so that trees built on different data can be averaged. seqhesion also summarises each pair over the trees that hold it, but differs in three ways. Its summary is the median, over those trees, of the diameter of the smallest clade holding both tips: a level, robust to a minority of discordant trees, rather than a mean of path lengths. Pairs no tree holds are left out of the linkage altogether. And the result is a hierarchy of groups with their evidence, not one resolved tree.

How far a combined tree departs from its inputs is often measured by the Robinson–Foulds (RF) distance (Robinson and Foulds, 1981): the number of clades found in one tree but not the other. RF supertrees (Bansal et al., 2010) minimise its total over the input trees. As an objective it does not suit this setting. It weighs every clade equally, so the many shallow splits that differ by chance between shards dominate it; it counts a polytomy as disagreement; and it yields no levels. As a measure of fit, seqhesion keeps a refined form of it. Each group's per-shard verdicts (Section 3.6) say whether a shard has the same clade, leaves it unresolved, or contradicts it. A shard's conflicts, summed over the groups among its tips, approximate its RF distance to the hierarchy without counting a lack of resolution as conflict.

**Consensus clustering.** The co-association matrix of evidence-accumulation clustering (Fred and Jain, 2005) records, for each pair of objects, how often many base clusterings put them together. Hierarchical clustering of that matrix then gives the final structure. Cluster ensembles (Strehl and Ghosh, 2002) include the case where each base clusterer sees only a subset of the objects. Consensus clustering (Monti et al., 2003) normalises co-assignment by co-sampling: a pair's score is the number of resamples that put it together divided by the number that contain both. It also defines per-item and per-cluster consensus scores. Minipatch consensus clustering (Gan and Allen, 2022) extends this to tiny subsamples and adaptive sampling. seqhesion's co-association is of this kind. Its base "clusterings" are shard trees, which hold neighbourhoods rather than random subsets. Each tree contributes a continuous join level per pair rather than a binary co-assignment, and pairs no shard holds are excluded from the linkage, not imputed.

**Support and resampling.** Felsenstein's bootstrap (Felsenstein, 1985), an application of Efron (1979), measures the stability of splits under resampling of alignment columns. The transfer bootstrap (Lemoine et al., 2018) relaxes it for large trees. Taxon jackknifing (Lanyon, 1985) perturbs the taxon set instead. seqhesion's overlapping shards are closer to a taxon jackknife combined with a re-run of alignment and inference. We validate the confidence measures by bootstrap resampling of whole shard trees (Section 5.2).

**Phylogenetic placement.** Placement tools such as pplacer (Matsen et al., 2010) and EPA-ng (Barbera et al., 2019) attach query sequences to a fixed reference tree without changing it. seqhesion places partial (ITS2-only) sequences in the same spirit. It adds them to stored shard alignments with MAFFT (Katoh and Frith, 2012) and re-infers the shard tree, reading the query's position from the re-inference while the stored trees, and so the hierarchy, stay unchanged (Section 3.10).

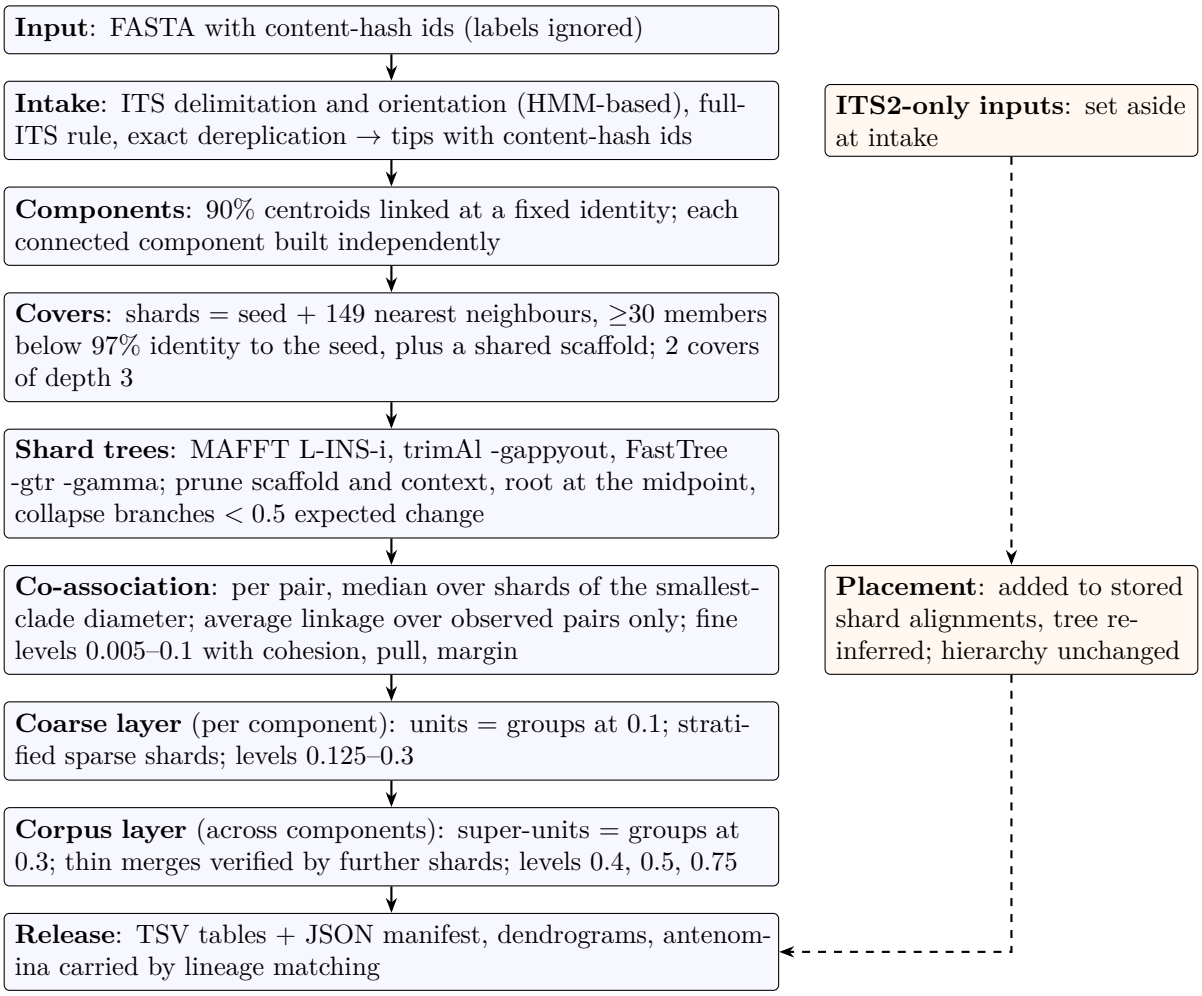



**Figure 1:** The seqhesion pipeline. Sequence identity (vsearch) is used only to decide what is sampled together: components, shard neighbourhoods, the scaffold and candidate shards for placement. Every grouping decision is made from tree distances in the shard trees.

# 3 Method

Figure 1 summarises the pipeline. Table 1 lists its main parameters. Two rules run through all of it:

- **Label-blind.** No step reads a name.
- **Tree distance and identity distance are not comparable.** Alignment identity (vsearch global identity) is used only to choose which sequences are sampled together. Every grouping and placement decision comes from tree distances: model-corrected, on trimmed alignment columns, with sub-resolution branches collapsed.

## 3.1 Intake

Input sequences are identified by content hashes. An input's id is the smaller of the SHA-256 digests of the cleaned sequence and of its IUPAC-aware reverse complement, so the id does not depend on orientation. seqhesion recomputes every id and refuses the whole input on any mismatch or duplicate. Anything after the id on a FASTA header is ignored.

The ITS subregions are delimited and the sequence oriented with pyitsx (Walker, 2026), an HMM-based ITS extractor in the manner of ITSx (Bengtsson-Palme et al., 2013). A sequence counts as *full ITS* when ITS1, 5.8S and ITS2 are all long enough (Table 1) and each flanking ribosomal anchor is either present or, when missing, the spacer beside it is long enough that it was probably trimmed at the boundary rather than truncated. Sequences with an LSU anchor and a usable ITS2 but an incomplete ITS1 or 5.8S are classed *ITS2-only*. They do not shape the hierarchy and are

**Table 1:** Main parameters of the current release.

| Step | Setting |
|---|---|
| Intake | full ITS: ITS1 $\geq$ 50, 5.8S $\geq$ 140, ITS2 $\geq$ 50 bp; an unanchored flanking spacer $\geq$ 150 bp; extracted full ITS $\leq$ 3,000 bp |
| Components | vsearch 90% centroids, linked at 86% centroid identity (production setting) |
| Neighbourhoods | vsearch global search, identity floor 80% (60% for small components, corpus-wide) |
| Shards | 150 sequences (seed + 149 nearest); top-up to $\geq$ 30 members below 97% identity to the seed; scaffold = 30 most abundant 85% centroids |
| Covers | 2 independent covers, each sequence in $\geq$ 3 shards per cover |
| Trees | `mafft -localpair -maxiterate 1000 -thread 1`; `trimal -gappyout`; `FastTree -gtr -gamma -nt` |
| Tree preparation | prune scaffold and context; root the members' tree at its midpoint (coarse and corpus shards: on their outgroup or context, then prune); collapse internal branches $< 0.5/W$ ($W$ = trimmed alignment width) |
| Fine levels | 0.005, 0.01, 0.015, 0.02, 0.03, 0.05, 0.075, 0.1 (substitutions/site, clade diameter) |
| Coarse layer | units at 0.1; 150 units per sparse shard; $\approx$24 expected shards per unit pair; 8 outgroup sequences; levels 0.125, 0.15, 0.2, 0.25, 0.3 |
| Corpus layer | super-units at 0.3; 3 representatives per unit, neighbour identity floor 50%; 150 units per shard + 20 context units; 2 covers of depth 6; levels 0.4, 0.5, 0.75; merges on $< 3$ votes verified by 8 more shards per round, until none remain; unit pairs on $< 3$ votes unobserved |
| Antenomina | mutual best match by Jaccard over shared inputs, $J > 1/2$ |

placed on it afterwards (Section 3.10). Chimeric, unreadable, overlong (over 3,000 bp; real full ITS tops out near 2,500 bp) and otherwise incomplete sequences are dropped and counted by class. Full-ITS extracts are exactly dereplicated. Each distinct extract is a *tip*, with an id derived from a hash of its sequence. Nothing is corrected: homopolymer errors, ambiguity codes and intragenomic variants are kept and their effects reported downstream.

### 3.2 Components

The dereplicated tips are clustered into 90% centroids (vsearch `-cluster_fast` (Rognes et al., 2016)), and centroids are linked when they match at or above a fixed identity (86% in the production build). Each connected component of this centroid graph is built independently. The centroids only locate components. They never define groups.

Many components are small. In the production corpus, 44% of tips lie in components of fewer than 150 tips, the size of one shard. Their tips also draw their nearest neighbours from the whole corpus (identity floor 60%). These outside sequences were introduced as rooting context; since local trees are rooted at the members' midpoint (Section 3.4) they serve only as context in the alignment and tree. They are pruned before rooting and co-association, and never grouped. A lab test found that 0% of small-component tips joined another component's tips at levels up to 0.075, so the pruning loses no grouping evidence. Singletons and components with no usable neighbour get no tree.

### 3.3 Shards and covers

A *shard* is a seed tip plus its 149 nearest neighbours by vsearch global identity. A *cover* is a set of shards in which every tip lies in at least *depth* shards (depth 3). Seeds are chosen greedily from the least-covered tips, and a random stream only breaks ties, so different streams give different but equally valid covers. No seed or member set is used twice within a cover. Tips with too few neighbours to seed a full shard receive smaller shards made of whatever neighbours they have, and every tip is accounted for as covered, under-covered or isolated.

Two refinements matter in practice.

- **Context quota.** In a dense species complex the 149 nearest neighbours may all be near-identical to the seed, so the shard never shows the species against outsiders. Each shard is therefore topped

up so that at least 30 of its members are below 97% identity to the seed. Identity here decides only what is *sampled*.

- **Scaffold.** Every shard also contains a fixed scaffold: the 30 most abundant 85% centroids of the component. It is pruned before rooting and before any co-association is counted, since it sits in every shard and would otherwise link everything. (It was introduced to provide rooting candidates; see Section 3.4.)

Two covers are drawn independently. In dense neighbourhoods they often draw exactly the same member set; across the large components 55–61% of cover-1 shards were such *twins* of a cover-0 shard. A twin is the same sample and gives the same tree, so it is built once and counted once.

### 3.4 Shard trees

Each shard is aligned with MAFFT L-INS-i (Katoh et al., 2005; Katoh and Standley, 2013), run single-threaded for determinism. The alignment is trimmed with trimAl `-gappyout` (Capella-Gutiérrez et al., 2009), and a tree is inferred with FastTree under GTR with gamma-distributed rates (Price et al., 2010). Each tree is then prepared as follows.

1. The scaffold and any context sequences are pruned.
2. The members' own tree is rooted at its midpoint (a tree of two members is left as it is).
3. Internal branches shorter than $0.5/W$ substitutions per site are collapsed into polytomies. Here $W$ is the trimmed alignment width, so the threshold is about half an expected change. Branches this short carry no resolvable signal, and FastTree resolves them arbitrarily.

The root matters because the join level of Section 3.5 is the diameter of the smallest clade holding a pair, and clades depend on the root. Releases up to 17 October 2026 rooted each tree on the scaffold tip farthest from the members, before pruning. In a large component the scaffold is the component's own most abundant centroids, sometimes only one or two. The root then fell inside a dense cluster of near-identical sequences, which became a grade with its relatives nested inside it, and pairs across the root joined at nearly the whole tree's width. Outgroups from outside the component failed in a different way: a distant outgroup attached inside a dense cluster. Midpoint rooting of the members' own tree avoids both (Section 5.3). Coarse-layer and corpus shards hold one member per unit, so no dense cluster can capture their root; they keep rooting on their outgroup or context sequences, which are then pruned.

### 3.5 Co-association and the fine hierarchy

Let $S$ be the set of distinct prepared shard trees of a component, pooled over both covers with twins counted once. For a shard $s$ holding tips $i$ and $j$, let $c_s(i,j)$ be the smallest clade of $s$ containing both, and define the shard's *join level*

$$d_s(i,j) = \operatorname{diam}(c_s(i,j)) = \max_{a,b \in c_s(i,j)} p_s(a,b), \tag{1}$$

where $p_s$ is the patristic distance in $s$. The diameter scale makes a level a statement about the widest pair inside a group, not about its height above a root.

**The "identical" join.** At a polytomy, plain diameters put every pair of the polytomy's children at the polytomy's full width. This split near-identical tips between groups. Following the view that a polytomy is missing information rather than conflict (Page, 2002), two tips in different children of a polytomy join at the diameter of just those two children, *but only if* that span is at most one expected change ($1/W$), i.e. when the data cannot tell them apart. Every other pair in the polytomy joins at its full width. A variant that applied the narrower span to all polytomy pairs grouped distinguishable lineages by noise-level distances, and in simulation it doubled the share of wrong groups at level 0.03 (4.4% to 8.6%).

**Pooling.** A pair $(i, j)$ is *observed* if at least one shard holds both. Its consensus join level is the median over the shards that hold it:

$$D(i,j) = \text{median}\{\, d_s(i,j) : s \in S,\ i,j \in s \,\}. \tag{2}$$

Neighbourhood shards observe under a tenth of all pairs in a large component (9.7% in one of 3,665 tips), mostly the near ones. An unobserved pair is not evidence that two tips are far apart; it is no evidence at all.

**Observed-pair average linkage.** The hierarchy is built by average linkage (Sokal and Michener, 1958) over observed pairs only:

$$D(A,B) = \frac{1}{|O_{AB}|} \sum_{(a,b)\in O_{AB}} D(a,b), \tag{3}$$

where $O_{AB}$ is the set of observed pairs between clusters $A$ and $B$. The update after a merge is the observed-pair-weighted mean of the parents' distances, so, as in ordinary average linkage, merge heights never decrease. Two clusters with no observed pair between them are never merged on evidence. Parts that nothing connects stay a forest, and are joined only formally at an infinite height for bookkeeping. Each merge records how many of its cross pairs were observed, so a join made on thin evidence is visible. Ties are broken by index, so the result depends only on the input pairs and not on their order.

The dendrogram is cut at eight *fine levels*, 0.005–0.1 substitutions per site (Table 1). A *group* at level $\ell$ is a cluster of two or more tips at that cut. A *node* is a distinct tip set over a run of consecutive levels. The levels are fixed and absolute, not adaptive. Several attempts to choose a species-level cut adaptively failed on real data. The hierarchy therefore reports every level and leaves the choice of level, clade by clade, to curators.

### 3.6 Evidence and confidence per group

Every shard's verdict on a pair is one *vote*. A shard $s$ holding $i$ and $j$ votes that the pair joins at level $\ell$ if $d_s(i,j) \le \ell$. All shares below are pooled over votes, so a pair seen by twenty shards weighs twenty times one seen by a single shard.

**Cohesion.** For a group $G$ at level $\ell$, cohesion is the share of votes on $G$'s internal pairs that join the pair at or below $\ell$:

$$\text{coh}(G) = \frac{\sum_{i<j\in G} \sum_{s\ni i,j} \mathbf{1}[d_s(i,j) \le \ell]}{\sum_{i<j\in G} |\{s : i,j \in s\}|}. \tag{4}$$

This is the analogue of the cluster consensus of Monti et al. (2003), with co-sampling as the denominator.

**Pull.** For each member $i$ of $G$, its *strongest outside target* $t(i)$ is the outside group (or ungrouped tip) whose pairs with $i$ have the highest pooled share of joining votes at $\ell$. The group's pull pools these over members:

$$\text{pull}(G) = \frac{\sum_{i\in G} h_i}{\sum_{i\in G} v_i}, \tag{5}$$

where $v_i$ is the number of votes on pairs between $i$ and $t(i)$, and $h_i$ is how many of them join at or below $\ell$. An earlier version used the maximum over members of each member's pull. That saturated in large groups: a member's pull often rests on a handful of votes ($3/3 = 1$), so the maximum reached 1 in most groups of 200 or more tips.

**Margin.** The confidence measure is

$$\text{margin}(G) = \text{coh}(G) - \text{pull}(G), \tag{6}$$

how much more strongly the shards join the group's own pairs than they join its members to anything outside. Section 5.2 shows that it predicts recurrence under resampling.

**Further annotations.** Each group also carries:

- *held*: the share of its pairs that some shard observes;
- per-shard *verdicts* for every shard holding two or more of its tips: *clade*, *unresolved* (a polytomy holds them with outsiders), *conflict* (a clade of the shard takes some of them together with an outsider), or *whole* (the shard holds no outsider and cannot tell);
- the stem length when it is a clade;
- its *spread* (widest and median consensus patristic distance between its tips);
- its nearest and second-nearest outside groups, with the *gap* ratio of nearest distance to spread;
- the minimum and median pairwise vsearch identity of its members, for reporting only.

Per tip and level, seqhesion reports membership (the share of joining votes on its pairs with the rest of its group) and its strongest outside pull. The design principle is to annotate rather than threshold. No group is hidden for low confidence, and a partition into units is a view that a consumer derives from these annotations.

**Stem nodes.** Fixed levels miss a clade that lives wholly between two of them: in the current release a 245-sequence clade of *Trametes versicolor* exists only between heights 0.021 and 0.026, so the levels 0.02 and 0.03 see it as two groups and then as part of a larger one. Every node of the hierarchy below 0.1 that is a group at no level, but whose stem (its parent's height minus its own) is at least 0.001, is therefore reported as well, as a *stem node* with the same cohesion, pull and margin, measured at a point inside its life. A longer minimum stem did not make these ids more stable: under bootstrap resampling of the shard trees, 75% of stem-node ids carried over against 90% of level groups', at thresholds from 0.001 to 0.0025. The current release has 17,557 stem nodes beside about 56,000 groups; about half of them are clusters of near-identical sequences below the finest level.

### 3.7 The coarse layer within a component

Above about 0.1, neighbourhood shards rarely hold two distant tips, so coarse merges would rest on the few near pairs that local shards happen to observe. In the main *Cortinariaceae* component, average linkage on local shards alone chained to 5 groups at 0.2 and 1 at 0.3. seqhesion therefore builds a separate layer above the fine hierarchy.

- **Units** are the fine groups at 0.1 and the tips that are in no group there.
- **Sparse shards** are stratified. Each holds one random member of each of 150 randomly chosen units, so every unit pair is equally likely to meet. Each also holds a fixed outgroup of 8 sequences from outside the component, used for rooting and then pruned (one member per unit leaves no dense cluster for the root to fall into). Enough shards are drawn that each unit pair is expected in about 24 of them. They are aligned, trimmed, inferred and prepared exactly like local shards.
- **The layer.** Each sparse shard's join level for a pair of tips in different units is one vote on that unit pair. The median over its votes is the unit pair's level. Average linkage over observed unit pairs gives a hierarchy of units, cut at 0.125–0.3, with cohesion, pull and margin computed over unit-pair votes.

Why a separate layer rather than one hierarchy? Feeding the sparse votes into the tip-level hierarchy leaves the fine levels nearly unchanged. But the far more numerous near pairs from local shards then dominate observed-pair averages, and the coarse levels chained (46 groups at 0.2, 11 at 0.3). The layer weighs unit pairs equally. Random rather than stratified sparse shards observed only a third of unit pairs at the same cost.

Components smaller than one shard need no sparse shards: every local shard already holds all their tips, so their unit layer is built from the local trees. A fine group that is still a single unit at a coarse level is the same node continuing, with the same identifier. The released dendrogram is the fine linkage inside each unit plus the layer's linkage above, so every group at every level is a clade of it.

### 3.8 The corpus layer across components

Components are cut apart at the centroid-graph threshold, so nothing in the fine or coarse hierarchies relates them. The corpus layer does, with the same machinery one level up.

- **Super-units** are each component's groups at 0.3 and its tips in none: about 4,800 over the production corpus.
- **Neighbours.** Each super-unit has up to three representatives (its tips with the most inputs). Units are ranked as neighbours by the best vsearch identity between representatives (floor 50%). As everywhere, identity only chooses what is sampled together.
- **Cover.** A cover over units is planned exactly as a cover over tips: each shard is a seed unit and its 149 nearest units, with 2 covers of depth 6. Units within reach meet in many shards, and distant ones, whose ITS is unalignable, never do. A shard holds one random member of each of its units. One random member of each of the seed's next 20 units is added as rooting context and pruned afterwards.
- **Layer.** Unit-pair median join levels and observed-pair average linkage, as in the coarse layer, cut at 0.4, 0.5 and 0.75. Units nothing connects stay a forest.
- **Verification.** At corpus distances representative identity barely ranks relatives, so some unit pairs meet in only one or two trees, by chance, and a merge can rest on such a pair. Every merge at or below 0.75 whose cross unit pairs hold fewer than 3 votes in all is therefore verified: up to 4 of its cross unit pairs go into 8 more shards, filled with random corpus units as context that is pruned before votes, so only the target pairs vote. The layer is rebuilt, and rounds repeat until no merge is thin. A unit pair still on fewer than 3 votes then counts as unobserved.
- **Weakest link.** Every coarse and corpus group reports the votes behind its own merge and the smallest such count in its subtree, so a group resting on a thin join anywhere below it is visible.

The levels stop at 0.75 because two independent draws stopped agreeing above it (Section 5.5). A corpus group can span several components; a component's top group that stays alone at a corpus level continues with the same identifier.

### 3.9 Antenomina: identities across releases

An *antenomen* is a group with an identity that outlives a single release, so that comments and curation can attach to it before, and independently of, any name. Every distinct group (node) gets one. When a new release is made, its groups are matched to the previous release's groups by their *inputs* (content-hash ids, which survive changes to extraction). The comparison is restricted to inputs present in both releases, so a group that merely gains new sequences is not penalised. A new group $N$ inherits the id of an old group $P$ if and only if

- each is the other's best match by Jaccard similarity (Jaccard, 1912), and
- $J(N, P) > 1/2$, i.e. most of each is the other.

On a tie in Jaccard, the candidate with fewer inputs outside the comparison wins. This matters when a group and its parent at a coarser level differ only by inputs the other release holds in no group. Every other group gets a newly minted id. Ids are eight Crockford base-32 characters shown as `XXXX-XXXX`. They look random but are reproducible: a hash of the release's input fingerprint and the group's inputs, lengthened on a collision with any id ever minted. Ids are never reused. Every comparison is logged as a lineage event (continued, born or retired). Deciding what happens to a curator's annotation on a group that split or merged is left to the curation tool.

### 3.10 Placement of partial sequences

ITS2-only inputs never shape the hierarchy. They are placed onto it through the stored shard trees:

1. **Candidate.** The query's ITS2 is compared with every tip's ITS2 (vsearch), and the closest tip is the candidate. This search only chooses where to look.
2. **Shards.** The candidate's three most central shards in each cover are used (the shard it seeds first, then by its neighbour rank), skipping twins.
3. **Re-inference.** In each shard the query is added to the stored L-INS-i alignment with `mafft -addfragments -keeplength` (Katoh and Frith, 2012), the shard's own trimAl column mask is applied, and a FastTree tree is re-inferred. The query's join levels with every member are read from the re-inferred tree. The stored tree, and so every existing pair's evidence, is untouched.
4. **Assignment.** The query's join level with each member is the median over shards. The query is assigned to the finest level at which some group's average join level to it is within the level, that group, and its ancestors.

As a diagnostic, `seqhesion` records whether a re-inference rearranged the query's surroundings relative to the stored tree. The same path, with `-add`, places new full-ITS sequences incrementally (Section 5.6). In production the re-inferences are planned once (one job per query and shard) and run as independent chunks; the release then reduces their join levels to placements.

## 4 Implementation and reproducibility

`seqhesion` is a Python package (NumPy (Harris et al., 2020), SciPy (Virtanen et al., 2020), ETE 3 (Huerta-Cepas et al., 2016)) that calls vsearch, MAFFT, trimAl and FastTree, plus an HMM-based ITS extractor. A command-line interface runs intake, component splitting, cover planning, tree building, hierarchy construction, the coarse and corpus layers, and release writing.

**Determinism and caching.** Every shard tree is cached under a key that fingerprints every input of that tree: the member sequences and their order, the recipe, and the tool identities. A hierarchy records the digests of what it was built from, and a stale artefact is refused rather than believed. MAFFT runs with `-thread 1`, so one invocation always gives one result, and shards with identical input are built once.

**Cross-platform equivalence.** Floating-point contraction changes MAFFT's output. On Linux, builds compiled with clang and fused multiply-add contraction produced alignments byte-identical to the macOS reference on every input tested. GCC builds reporting the same version string differed on about a third of shards. Cache keys therefore never trust a version string alone, and a canary set of alignments is checked before use. FastTree builds differed between platforms only in how they resolved ties. After the branch-collapse step, 72% of trees were identical in topology, 20% differed only in tie resolution, and 8% were sensitive. End to end this kept 98% of antenomina, against 89% for merely redrawing the covers, so the platform effect is below the method's own resampling noise. Alignments are shared across platforms. Trees are kept per platform.

**Compute and scale.** Almost all of the cost is shard trees, and almost all of a tree's cost is its L-INS-i alignment; FastTree takes about a second. Shards have a fixed size (150–210 sequences with the scaffold), so the cost of one tree is bounded and the number of trees grows linearly with the corpus. No step aligns or infers a tree over a whole component. A single L-INS-i alignment grows roughly with the cube of its size and already takes 36 minutes at 2,000 sequences on our hardware; the largest component holds 13,098 tips. Shard trees are independent, so they parallelise trivially. Table 2 counts the trees behind the current release.

Most trees were built on AWS Batch, on c7a.xlarge instances with four cores each. The fine covers took 23.1 instance-hours for 30,571 shards, the sparse trees 0.44, and the corpus trees 13.8 for two independent plans (the second served the stability check of Section 5.5). A corpus tree costs about three times as much as a fine shard (about 37 against 11 core-seconds), because its

**Table 2:** Shard trees behind release 20261006.01f.

| Stage | Trees | Sequences per tree (median) |
|---|---|---|
| Fine covers: shards | 36,386 | 158 |
| distinct trees (cross-cover twins built once) | 25,618 | |
| Coarse layer: sparse trees | 3,119 | 22 |
| Corpus layer: trees (one plan) | 2,677 | 170 |
| Distinct trees in the release | 31,414 | |

170 sequences are divergent. Scaled to every shard, a build from an empty cache takes about 35 instance-hours, or 140 core-hours. That is about 3.4 core-seconds per tip, about 4.5 hours of wall time on eight instances, and under US$10 of on-demand compute. The remaining steps run on one workstation: the hierarchy of the largest component (1,616 distinct trees) takes under five minutes, and writing the full release about 13 minutes. The cache holds about 560 KB per tree. Because trees are cached by content, a later build only builds shards whose members changed. Placing the 18,583 placeable ITS2-only inputs of the current release took 100,347 shard re-inferences, 11.5 job-hours on the same instances.

**Optimised MAFFT.** Because alignment dominates, a faster MAFFT pays off directly. We used optimised builds of MAFFT 7.526 that are required to give byte-identical output to stock MAFFT for our invocation, verified on canary sets of real shards before each was adopted. Their source and a technical report are public (Claude Code (Claude Opus 5.5, Anthropic) and Walker, 2026). On the Apple M1 workstation, the build used here runs L-INS-i 6.5 times faster than stock in a single run and 4.6 times faster with eight alignments at a time, the production setting. On AWS, a build that ports the vector kernels to AVX-512 runs 5.1–5.9 times faster than stock with four alignments at a time. Choosing c7a instances, with four physical cores, over c7i, with two hyperthreaded cores, gained a further factor of about 1.7 (16.0 against 26.9 seconds per shard). With stock MAFFT the same build would have needed roughly five times the core-hours. Because the output is byte-identical, the optimised builds share the tree cache with stock MAFFT, and the choice of build never changes a result. The work behind them also found that stock MAFFT itself is not identical across platforms: whether the compiler fuses multiply-adds changed about a third of alignments, which is why the reference on Linux is a clang build with fused multiply-adds (see Cross-platform equivalence above).

**Releases.** A release is a set of TSV tables and a JSON manifest with a versioned schema (currently 0.7). It contains:

- every input and its status: built, not built, or dropped with its intake class;
- tips with their extracted sequences;
- groups with their evidence, and per-level evidence;
- group membership, and per-tip membership and pull at each level;
- ITS2 placements: every ITS2-only input, placed (its group, level and join) or why not;
- for every coarse and corpus group, the votes behind its merge and its weakest merge;
- the lineage log and a register of every id ever minted;
- Newick dendrograms (per component, and one corpus-wide tree in which every group at every level is a clade);
- the prepared shard trees themselves, so consumers can recompute any distance.

No table contains a label.

## 5 Results

All results in this section come from in-house (“lab”) experiments run during development, mostly on one production corpus. They are internal measurements, not peer-reviewed benchmarks. Name

**Table 3:** Scale of the current production build.

| | |
|---|---|
| Input sequences | 225,985 |
| ITS2-only (not built; placed on the hierarchy) | 19,589 (18,580 placed) |
| Distinct full-ITS tips | 146,946 |
| Components built | 4,174 |
| Fine groups (levels 0.005–0.1) | 51,269 |
| Stem nodes (between levels) | 17,767 |
| Coarse groups (levels 0.125–0.3) | 3,917 |
| Super-units for the corpus layer | 4,769 |
| Corpus groups (levels 0.4–0.75) | 1,603 |

**Table 4:** AUC for flagging groups that recur in few bootstrap resamples of the shard trees (334 components, 20 resamples each; one component 5). At 101+ tips no group recurred in under half the resamples, so only the $< 95\%$ column is defined there.

| Measure | recur $< 50\%$ | recur $< 95\%$ | recur $< 95\%$, groups of 101+ tips |
|---|---|---|---|
| margin = cohesion − pooled pull | 0.94 | 0.87 | 0.83 |
| cohesion − max-over-members pull | 0.92 | 0.85 | 0.69 |
| cohesion alone | 0.80 | 0.72 | 0.61 |

comparisons appear only as sanity checks: no parameter was tuned to them. Except where Section 5.3 says otherwise, the experiments predate midpoint rooting.

### 5.1 Production corpus

Table 3 summarises the current corpus (release 20261018.01f). The largest component, a *Cortinariaceae* component of 8,806 tips, is the test bed for the coarse layer below. In the release with pooled covers and pooled pull, negative margins were rare: 0.6%, 0.2%, 0.2% and 0% of groups of 2–5, 6–20, 21–199 and 200+ tips.

### 5.2 Confidence predicts recurrence under resampling

To test whether the per-group evidence means anything, we bootstrapped the shard trees. In each of 20 resamples, a component's distinct trees were drawn with replacement (Efron, 1979), and the consensus levels and hierarchy were rebuilt from them alone. A group *recurs* in a resample if the resample has a group at the same level that is its mutual best match with $J > 1/2$ (the antenomen rule). Over 334 components (25.6K group-levels), 80–93% of groups recurred in at least 95% of resamples and 1–4% in fewer than half. Pairs were the least stable group size (4.7% below one half).

Table 4 gives each measure's ability to flag groups that recur rarely, as the area under the ROC curve (Hanley and McNeil, 1982). Overall the margin ties with the best candidate tried (a median over members, within 0.003), and in groups of 101+ tips it is the best (0.83). The maximum-over-members pull it replaced scored well overall only because of small groups. A fixed split-half agreement between the two covers (the former "replication" measure) did *not* predict recurrence: groups it flagged recurred at the same mean rate as the rest (0.960 vs 0.966). It was dropped in favour of pooling both covers.

### 5.3 Rooting of local shards

Curators asked why some groups of near-identical sequences formed only at 0.125. In a 489-tip *Hygrocybe* component with a one-tip scaffold, the root fell inside the near-identical cluster in 75 of 76 trees, and its cross-root pairs joined at a median of 0.124 against a patristic distance of 0.008. The same trees rooted at the midpoint gave 0.030; nothing else moved. We compared both roots on the existing trees of seven large components: four with such reports, a 775-tip *Trametes* component,

**Table 5:** Agreement (ARI over tips) between two independent draws of sparse shards, main *Cortinariaceae* component.

| Sparse trees per draw | 0.15 | 0.25 | 0.3 | 0.4 |
|---|---|---|---|---|
| 26 (≈6 votes per unit pair) | 0.93 | 0.81 | 0.56 | – |
| 102 (≈24 votes per unit pair) | 0.98 | 0.86 | 0.87 | 0.85 |

and two controls with five or more scaffold tips. Midpoint rooting cut the coarse-layer merges below 0.1 in the four reported components from 48, 31, 23 and 14 to 1, 0, 3 and 0, and three of the four groups now form at 0.02–0.05. The fourth is only partly mended: 90 of its 146 tips share a group at 0.1 (53 before), but the whole still forms at 0.125. The controls were unchanged (ARI about 1.0), as was *Trametes* (ARI $\geq$ 0.99 at every level but 0.1, at 0.69). Rooting on 8 outgroups from outside the component instead also mended the four reported cases, but in *Trametes* the outgroup attached inside a dense cluster in all 102 trees and fragmented it (24 to 72 groups at 0.05). Across the corpus, coarse-layer merges below 0.1 in large components fell from 855 to 519 with midpoint rooting. Not all of these are rooting: sparse shards sample a component differently from local ones.

### 5.4 The coarse layer

On the 8,806-tip *Cortinariaceae* component (308 units at 0.1), we drew independent sets of sparse shards and compared the resulting layers by the adjusted Rand index (ARI) over tips (Hubert and Arabie, 1985) (Table 5). With 26 trees (about 6 votes per unit pair) the layers agreed well at 0.15 but poorly at 0.3. With 102 trees (about 24 votes) agreement held to 0.4. More sampling extends the stable range. The production layer for this component, built separately, agreed with the lab draws at ARI 0.83–0.97 up to 0.25 and 0.62–0.70 at 0.3.

**Sanity check against names.** For each genus in the component (the genera of the revised classification of Liimatainen et al. (2022)) we took the best-matching node at any height, by F1 over the tips named in that genus. The names come from the downstream curation pipeline. Each record's standing name is read from the identifications made on it, with expert validators taking precedence over community identifications. Each clade of a cut of the fine hierarchy (levels $\leq$ 0.1) then takes the plurality name of its records. The genus, subgenus, section and family of that name are read from iNaturalist's taxonomy. These are validators' working names, about half of them provisional codes, not determinations. Because a tip's name is that of its fine-level clade, the check asks whether coarse nodes bring together the fine clades of one genus, not whether single tips agree. Mean genus F1 was 0.91–0.92 with the sparse layer, against 0.82 with local shards alone. For example, *Phlegmacium* rose from 0.78 to 0.97 and *Calonarius* from 0.82 to 0.99. Genera formed at different heights (about 0.12–0.27), so no single cut reproduces them, which is consistent with curators choosing levels clade by clade. *Thaxterogaster* stayed at F1 0.62–0.69 with about 24 votes per unit pair, and with nested shards inside its coarse group added (0.44 with 26 trees, 0.46 with local shards alone). Its internal depth (about 0.26) is about equal to its distance to neighbouring genera (0.26–0.28), within the spread of the votes: an ITS backbone near-tie, not a sampling limit.

### 5.5 The corpus layer

In a label-blind experiment, we selected the 122 components with any tip within 75% identity of the main *Cortinariaceae* component (39,349 tips, 318 super-units) and built two independent draws of about 109 sparse trees each. The draws agreed at ARI 0.90, 0.95, 0.98 and 0.93 at levels 0.3, 0.4, 0.5 and 0.75, but only 0.66 at 1.0, and above 1.5 everything merged into one group. Families came out as single nodes: the best-node F1 for families averaged 0.96–0.97, against 0.82 with components kept apart. Hymenogastraceae, spread over 11 components, rose from 0.51 to 0.99, and Strophariaceae from 0.53 to 1.00. *Cortinariaceae* itself was the exception: F1 0.71 in one draw (at height 0.64) and

**Table 6:** Leave-one-out placement: tips (of 100) assigned to their own group at each level.

| Level | 0.005 | 0.01 | 0.015 | 0.02 | 0.03 | 0.05 | 0.075 | 0.1 |
|---|---|---|---|---|---|---|---|---|
| ITS2 via shard trees | 78 | 89 | 87 | 93 | 94 | 96 | 93 | 93 |
| Full ITS via shard trees | 98 | 96 | 97 | 98 | 99 | 99 | 98 | 97 |
| ITS2 via vsearch identity | 53 | 66 | 74 | 79 | 85 | 91 | 95 | 97 |

1.00 in the other only at 1.10, above the stable range. The production cover design, run on the same set, gave a family F1 of 0.97 and agreed with the experimental draws at ARI 0.70–0.88.

Over the whole corpus we built two independent production plans over all 4,813 super-units (2,677 and 2,658 shards; the release uses plan A). Each observed about 27% of all unit pairs, with a median of 4 votes per observed pair: the cover samples neighbourhoods of units, not all pairs.

| Corpus level | 0.4 | 0.5 | 0.75 |
|---|---|---|---|
| ARI between plans A and B | 0.935 | 0.943 | 0.931 |
| Corpus groups (2+ units), plan A (B) | 969 (990) | 827 (847) | 452 (452) |
| Groups of A recurring in B (lineage rule) | 91% | 88% | 77% |

The two plans agree about as well as the lab draws did, with fewer votes per pair, and recurrence falls with height as it should: the 0.75 level is the least certain and is reported as such through each group's margin (median 0.53, 0.49 and 0.41 at the three levels). As a sanity check only, over the 174 families with at least 30 tips the mean best-node F1 was 0.92 in both plans against 0.66 with components kept apart; over 446 genera, 0.89 against 0.76. In release 20261018.01f, 1,572 of the 1,603 corpus groups span more than one component.

**Verification.** Curators reported a few corpus groups that joined clades of different orders. Each rested on a bridge unit pair seen by one or two corpus trees. Deeper covers did not cure this: at depth 24 (4.6 times the trees) 41% of such pairs still had two votes or fewer. Over the whole corpus, verification rounds (Section 3.8) reduced thin merges from 157 to 84, 49, 12, 6, 1 and 0 over six rounds (124 extra trees), and the layer stayed close to the unverified one (ARI 0.995, 0.965 and 0.975 at 0.4, 0.5 and 0.75); every reported case came apart. In the current release the rounds went from 96 to 1 thin merge after six rounds, and that one did not clear in four more: a two-sequence component whose weak joins move to a new partner each round. With the 3-vote minimum it joins nothing below 0.95, so no merge at or below 0.75 in the release rests on fewer than 3 votes.

### 5.6 Placement of ITS2-only sequences

In a leave-one-out test on 100 tips of an 8,271-tip component, we cut each tip's ITS2 from its own sequence and placed it as if it were an ITS2-only input. We then checked whether it was assigned to the group the tip belongs to (Table 6). Through the shard trees, ITS2 alone landed in its own group 78–96% of the time across levels. The full ITS by the same path landed in its own group 96–99% of the time. Placement by vsearch identity, the previous method, mostly failed by declining to decide at fine levels. Wrong assignments (another group) by the tree path were 0–4 per level. Leave-one-out on tips is optimistic for real ITS2-only reads, which come from other amplicons with other error profiles.

**Incremental addition (experimental).** The full-ITS row is also a test of adding new sequences without a rebuild. In a further experiment, 308 new sequences (mostly type sequences) were placed into an existing 8,271-tip component this way and compared with a full rebuild. Placements that shared no previously present tips with the rebuild's group occurred 0–3 times per level. The incremental path took 1,848 shard re-inferences (13.5 minutes on 7 processes), against more than 3 hours for the rebuild. It is conservative at the finest level, because a placed sequence draws on 6 shards rather than the 10–14 a rebuilt tip sits in. Incremental releases are not yet part of production.

# 6 Limitations

**A navigation aid, not a phylogeny.** seqhesion's dendrograms are average-linkage summaries of median clade diameters from many local trees. They are not phylogenetic trees: branch lengths are differences of merge heights, and topology above the fine levels comes from sparse samples. The hierarchy is a starting point for research and a map for curators. Nobody should publish a phylogeny on this evidence alone.

**ITS does not resolve deep backbones.** Agreement between independent draws held to about 0.75 and failed by 1.0. Order-level and deeper relationships, and some genus-level backbones (*Thaxterogaster*, and *Cortinariaceae* as a whole), are near-ties on ITS. They need other loci, or a reference backbone, to resolve.

**Identity only for sampling.** Component boundaries, shard neighbourhoods and placement candidates are chosen by identity, which shapes what evidence exists. A pair no shard holds contributes nothing, so two clusters can stay apart for lack of evidence rather than because of evidence against them. The *held* and *observed* counts make this visible, but they do not remove it.

**Star-shaped clusters.** When a tree is a star around one central sequence, every pair across two arms of the star has the star as its smallest common clade, and joins at the star's full width. In one 103-tip component this put a group's formation at 0.13. Neither root choice changes it; it is a property of the clade-diameter join and remains open.

**Correlated evidence.** Overlapping shards share sequences, alignments and the inference pipeline. Bootstrap resampling of shard trees treats them as independent and so understates uncertainty. Systematic errors common to every shard (alignment of indel-rich regions, model misfit, an extraction error) are invisible to any measure built from those shards.

**Reproducibility is not correctness.** Agreement between covers or draws measures whether the method gives the same answer twice. It cannot judge whether that answer is right, and a change that biases both draws alike will look stable.

**Names are only a sanity check.** F1 against names is reported for orientation. Names in the corpus are themselves uncertain, and the taxonomy used for checking may lag behind the sequences. A low F1 may indicate a problem with either side.

**Granularity is left to curators.** seqhesion does not decide which level is "species". Fixed absolute levels are reported, and curators choose levels per clade. No context-free rule we tried picked a species-level cut reliably.

**Noise and intragenomic variation are reported, not corrected.** Consensus sequences are used as they are. Sequencing errors (notably homopolymer length errors in nanopore data) and intragenomic ITS paralogues (Bradshaw et al., 2023) can form or split fine groups. The method reports them through low margins, polytomies and nearest-group distances. It does not mask them.

**Release-to-release churn.** A full rebuild draws new covers. In a lab rebuild after adding only a few sequences, 10–13% of antenomina were retired, almost entirely from resampling rather than from the new data. Antenomina are designed to absorb this, but a group near a boundary may still change identity between releases. Append-only covers and incremental placement are being developed to reduce churn.

**Scope.** The method has been developed and tested on fungal full ITS, mostly from macrofungal collections, and on one large corpus. Its parameters (shard size, levels, quotas) have not been tested on other markers or on environmental amplicon data.

# 7 Availability

seqhesion is open source under the BSD 3-Clause licence. Source code: https://github.com/joshuaowalker/seqhesion; package `seqhesion` on PyPI (version 0.2.0). This paper is a companion to the repository and is revised there. It requires Python $\geq$ 3.10, NumPy, SciPy and ETE 3, pyitsx (Walker, 2026), an HMM-based ITS extractor, and the external tools vsearch, MAFFT, trimAl and FastTree.

# Acknowledgements

The sequences analysed here come mainly from MycoMap (its sequence pages, CSV export and API), whose records are mostly iNaturalist observations, plus MycoPortal and Mushroom Observer records; additional ITS sequences recorded on iNaturalist observations; and type sequences from GenBank. We thank the community scientists, sequencing laboratories and sequence validators whose vouchered collections and consensus sequences make up that corpus; the method exists to make their work easier to navigate.

**Author contributions.** J.W. conceived the project, set its principles and made the design decisions. Claude Code (Claude Opus 5.5, Anthropic) implemented the software, ran the experiments and drafted the text under J.W.'s direction. J.W. takes full responsibility for the content.

# References

Kessy Abarenkov, R. Henrik Nilsson, Karl-Henrik Larsson, Andy F. S. Taylor, Tom W. May, Tobias Guldberg Frøslev, Julia Pawlowska, Björn Lindahl, Kadri Põldmaa, Camille Truong, et al. The UNITE database for molecular identification and taxonomic communication of fungi and other eukaryotes: sequences, taxa and classifications reconsidered. *Nucleic Acids Research*, 52(D1):D791–D797, 2024. doi: 10.1093/nar/gkad1039.

Metin Balaban, Yueyu Jiang, Qiyun Zhu, Daniel McDonald, Rob Knight, and Siavash Mirarab. Generation of accurate, expandable phylogenomic trees with uDance. *Nature Biotechnology*, 42(5):768–777, 2024. doi: 10.1038/s41587-023-01868-8.

Mukul S. Bansal, J. Gordon Burleigh, Oliver Eulenstein, and David Fernández-Baca. Robinson-Foulds supertrees. *Algorithms for Molecular Biology*, 5:18, 2010. doi: 10.1186/1748-7188-5-18.

Pierre Barbera, Alexey M. Kozlov, Lucas Czech, Benoit Morel, Diego Darriba, Tomáš Flouri, and Alexandros Stamatakis. EPA-ng: Massively parallel evolutionary placement of genetic sequences. *Systematic Biology*, 68(2):365–369, 2019. doi: 10.1093/sysbio/syy054.

Johan Bengtsson-Palme, Martin Ryberg, Martin Hartmann, Sara Branco, Zheng Wang, Anna Godhe, Pierre De Wit, Marisol Sánchez-García, Ingo Ebersberger, Filipe de Sousa, et al. Improved software detection and extraction of ITS1 and ITS2 from ribosomal ITS sequences of fungi and other eukaryotes for analysis of environmental sequencing data. *Methods in Ecology and Evolution*, 4(10):914–919, 2013. doi: 10.1111/2041-210X.12073.

Michael J. Bradshaw, M. Catherine Aime, Antonis Rokas, Autumn Maust, Swarnalatha Moparthi, Keila Jellings, Alexander M. Pane, Dylan Hendricks, Binod Pandey, Yuanning Li, and Donald H. Pfister. Extensive intragenomic variation in the internal transcribed spacer region of fungi. *iScience*, 26(8):107317, 2023. doi: 10.1016/j.isci.2023.107317.

Salvador Capella-Gutiérrez, José M. Silla-Martínez, and Toni Gabaldón. trimAl: a tool for automated alignment trimming in large-scale phylogenetic analyses. *Bioinformatics*, 25(15):1972–1973, 2009. doi: 10.1093/bioinformatics/btp348.

Claude Code (Claude Opus 5.5, Anthropic) and Josh Walker. Exact, output-identical acceleration of MAFFT on Apple silicon and x86. Technical report, GitHub repository joshuaowalker/mafft, `paper/mafft-apple-silicon.pdf`, 2026. URL https://github.com/joshuaowalker/mafft.

Alexis Criscuolo, Vincent Berry, Emmanuel J. P. Douzery, and Olivier Gascuel. SDM: A fast distance-based approach for (super)tree building in phylogenomics. *Systematic Biology*, 55(5):740–755, 2006. doi: 10.1080/10635150600969872.

B. Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979. doi: 10.1214/aos/1176344552.

Joseph Felsenstein. Confidence limits on phylogenies: An approach using the bootstrap. *Evolution*, 39(4): 783–791, 1985. doi: 10.1111/j.1558-5646.1985.tb00420.x.

Markus Fleischauer and Sebastian Böcker. Collecting reliable clades using the Greedy Strict Consensus Merger. *PeerJ*, 4:e2172, 2016. doi: 10.7717/peerj.2172.

Ana L. N. Fred and Anil K. Jain. Combining multiple clusterings using evidence accumulation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 27(6):835–850, 2005. doi: 10.1109/TPAMI.2005.113.

Tomochika Fujisawa and Timothy G. Barraclough. Delimiting species using single-locus data and the Generalized Mixed Yule Coalescent approach: A revised method and evaluation on simulated data sets. *Systematic Biology*, 62(5):707–724, 2013. doi: 10.1093/sysbio/syt033.

Luqin Gan and Genevera I. Allen. Fast and interpretable consensus clustering via minipatch learning. *PLoS Computational Biology*, 18(10):e1010577, 2022. doi: 10.1371/journal.pcbi.1010577.

Sigisfredo Garnica, Max Emil Schön, Kessy Abarenkov, Kai Riess, Kare Liimatainen, Tuula Niskanen, Bálint Dima, Karl Soop, Tobias Guldberg Frøslev, Thomas Stjernegaard Jeppesen, Ursula Peintner, Regina Kuhnert-Finkernagel, Tor Erik Brandrud, Günter Saar, Bernhard Oertel, and Joseph F. Ammirati. Determining threshold values for barcoding fungi: lessons from Cortinarius (Basidiomycota), a highly diverse and widespread ectomycorrhizal genus. *FEMS Microbiology Ecology*, 92(4):fiw045, 2016. doi: 10.1093/femsec/fiw045.

J. A. Hanley and B. J. McNeil. The meaning and use of the area under a receiver operating characteristic (ROC) curve. *Radiology*, 143(1):29–36, 1982. doi: 10.1148/radiology.143.1.7063747.

Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, Robert Kern, Matti Picus, et al. Array programming with NumPy. *Nature*, 585(7825):357–362, 2020. doi: 10.1038/s41586-020-2649-2.

Paul D. N. Hebert, Alina Cywinska, Shelley L. Ball, and Jeremy R. deWaard. Biological identifications through DNA barcodes. *Proceedings of the Royal Society of London. Series B: Biological Sciences*, 270 (1512):313–321, 2003. doi: 10.1098/rspb.2002.2218.

Lawrence Hubert and Phipps Arabie. Comparing partitions. *Journal of Classification*, 2(1):193–218, 1985. doi: 10.1007/BF01908075.

Jaime Huerta-Cepas, François Serra, and Peer Bork. ETE 3: Reconstruction, analysis, and visualization of phylogenomic data. *Molecular Biology and Evolution*, 33(6):1635–1638, 2016. doi: 10.1093/molbev/msw046.

Paul Jaccard. The distribution of the flora in the alpine zone. *New Phytologist*, 11(2):37–50, 1912. doi: 10.1111/j.1469-8137.1912.tb05611.x.

P. Kapli, S. Lutteropp, J. Zhang, K. Kobert, P. Pavlidis, A. Stamatakis, and T. Flouri. Multi-rate Poisson tree processes for single-locus species delimitation under maximum likelihood and Markov chain Monte Carlo. *Bioinformatics*, 33(11):1630–1638, 2017. doi: 10.1093/bioinformatics/btx025.

Kazutaka Katoh and Martin C. Frith. Adding unaligned sequences into an existing alignment using MAFFT and LAST. *Bioinformatics*, 28(23):3144–3146, 2012. doi: 10.1093/bioinformatics/bts578.

Kazutaka Katoh and Daron M. Standley. MAFFT multiple sequence alignment software version 7: Improvements in performance and usability. *Molecular Biology and Evolution*, 30(4):772–780, 2013. doi: 10.1093/molbev/mst010.

Kazutaka Katoh, Kei-ichi Kuma, Hiroyuki Toh, and Takashi Miyata. MAFFT version 5: improvement in accuracy of multiple sequence alignment. *Nucleic Acids Research*, 33(2):511–518, 2005. doi: 10.1093/nar/gki198.

Urmas Kõljalg, R. Henrik Nilsson, Kessy Abarenkov, Leho Tedersoo, Andy F. S. Taylor, Mohammad Bahram, Scott T. Bates, Thomas D. Bruns, Johan Bengtsson-Palme, Tony M. Callaghan, et al. Towards a unified paradigm for sequence-based identification of fungi. *Molecular Ecology*, 22(21):5271–5277, 2013. doi: 10.1111/mec.12481.

Scott M. Lanyon. Detecting internal inconsistencies in distance data. *Systematic Zoology*, 34(4):397–403, 1985. doi: 10.1093/sysbio/34.4.397.

François-Joseph Lapointe and Guy Cucumel. The average consensus procedure: Combination of weighted

trees containing identical or overlapping sets of taxa. *Systematic Biology*, 46(2):306–312, 1997. doi: 10.1093/sysbio/46.2.306.

F. Lemoine, J.-B. Domelevo Entfellner, E. Wilkinson, D. Correia, M. Dávila Felipe, T. De Oliveira, and O. Gascuel. Renewing Felsenstein's phylogenetic bootstrap in the era of big data. *Nature*, 556(7702): 452–456, 2018. doi: 10.1038/s41586-018-0043-0.

Kare Liimatainen, Jan T. Kim, Lisa Pokorny, Paul M. Kirk, Bryn Dentinger, and Tuula Niskanen. Taming the beast: a revised classification of Cortinariaceae based on genomic data. *Fungal Diversity*, 112(1):89–170, 2022. doi: 10.1007/s13225-022-00499-9.

Frederick A. Matsen, Robin B. Kodner, and E. Virginia Armbrust. pplacer: linear time maximum-likelihood and Bayesian phylogenetic placement of sequences onto a fixed reference tree. *BMC Bioinformatics*, 11:538, 2010. doi: 10.1186/1471-2105-11-538.

Robert N. McArthur, Ahad N. Zehmakan, Michael A. Charleston, Yu Lin, and Gavin Huttley. Spectral cluster supertree: Fast and statistically robust merging of rooted phylogenetic trees. *Frontiers in Molecular Biosciences*, 11:1432495, 2024. doi: 10.3389/fmolb.2024.1432495.

Daniel McDonald, Yueyu Jiang, Metin Balaban, Kalen Cantrell, Qiyun Zhu, Antonio Gonzalez, James T. Morton, Giorgia Nicolaou, Donovan H. Parks, Søren M. Karst, Mads Albertsen, Philip Hugenholtz, Todd DeSantis, Se Jin Song, Andrew Bartko, Aki S. Havulinna, Pekka Jousilahti, Susan Cheng, Michael Inouye, Teemu Niiranen, Mohit Jain, Veikko Salomaa, Leo Lahti, Siavash Mirarab, and Rob Knight. Greengenes2 unifies microbial data in a single reference tree. *Nature Biotechnology*, 42(5):715–718, 2024. doi: 10.1038/s41587-023-01845-1.

Christopher P. Meyer and Gustav Paulay. DNA barcoding: Error rates based on comprehensive sampling. *PLoS Biology*, 3(12):e422, 2005. doi: 10.1371/journal.pbio.0030422.

Siavash Mirarab, Nam Nguyen, Sheng Guo, Li-San Wang, Junhyong Kim, and Tandy Warnow. PASTA: Ultra-large multiple sequence alignment for nucleotide and amino-acid sequences. *Journal of Computational Biology*, 22(5):377–386, 2015. doi: 10.1089/cmb.2014.0156.

Stefano Monti, Pablo Tamayo, Jill Mesirov, and Todd Golub. Consensus clustering: A resampling-based method for class discovery and visualization of gene expression microarray data. *Machine Learning*, 52 (1–2):91–118, 2003. doi: 10.1023/A:1023949509487.

Serita Nelesen, Kevin Liu, Li-San Wang, C. Randal Linder, and Tandy Warnow. DACTAL: Divide-and-conquer trees (almost) without alignments. *Bioinformatics*, 28(12):i274–i282, 2012. doi: 10.1093/bioinformatics/bts218.

R. Henrik Nilsson, Erik Kristiansson, Martin Ryberg, Nils Hallenberg, and Karl-Henrik Larsson. Intraspecific ITS variability in the kingdom Fungi as expressed in the international sequence databases and its implications for molecular species identification. *Evolutionary Bioinformatics*, 4:193–201, 2008. doi: 10.4137/EBO.S653.

Roderic D. M. Page. Modified mincut supertrees. In *Algorithms in Bioinformatics (WABI 2002)*, volume 2452 of *Lecture Notes in Computer Science*, pages 537–551, Berlin, Heidelberg, 2002. Springer. doi: 10.1007/3-540-45784-4_41.

Joan Pons, Timothy G. Barraclough, Jesus Gomez-Zurita, Anabela Cardoso, Daniel P. Duran, Steaphan Hazell, Sophien Kamoun, William D. Sumlin, and Alfried P. Vogler. Sequence-based species delimitation for the DNA taxonomy of undescribed insects. *Systematic Biology*, 55(4):595–609, 2006. doi: 10.1080/10635150600852011.

Morgan N. Price, Paramvir S. Dehal, and Adam P. Arkin. FastTree 2 – approximately maximum-likelihood trees for large alignments. *PLoS ONE*, 5(3):e9490, 2010. doi: 10.1371/journal.pone.0009490.

N. Puillandre, A. Lambert, S. Brouillet, and G. Achaz. ABGD, Automatic Barcode Gap Discovery for primary species delimitation. *Molecular Ecology*, 21(8):1864–1877, 2012. doi: 10.1111/j.1365-294X.2011.05239.x.

Nicolas Puillandre, Sophie Brouillet, and Guillaume Achaz. ASAP: Assemble species by automatic partitioning. *Molecular Ecology Resources*, 21(2):609–620, 2021. doi: 10.1111/1755-0998.13281.

Sujeevan Ratnasingham and Paul D. N. Hebert. A DNA-based registry for all animal species: The barcode index number (BIN) system. *PLoS ONE*, 8(7):e66213, 2013. doi: 10.1371/journal.pone.0066213.

D. F. Robinson and L. R. Foulds. Comparison of phylogenetic trees. *Mathematical Biosciences*, 53(1–2): 131–147, 1981. doi: 10.1016/0025-5564(81)90043-2.

Torbjørn Rognes, Tomáš Flouri, Ben Nichols, Christopher Quince, and Frédéric Mahé. VSEARCH: a versatile open source tool for metagenomics. *PeerJ*, 4:e2584, 2016. doi: 10.7717/peerj.2584.

Conrad L. Schoch, Keith A. Seifert, Sabine Huhndorf, Vincent Robert, John L. Spouge, C. André Levesque, Wen Chen, Fungal Barcoding Consortium, et al. Nuclear ribosomal internal transcribed spacer (ITS) region as a universal DNA barcode marker for Fungi. *Proceedings of the National Academy of Sciences of the*

*United States of America*, 109(16):6241–6246, 2012. doi: 10.1073/pnas.1117018109. Fungal Barcoding Consortium; 158 authors listed in Crossref.

Charles Semple and Mike Steel. A supertree method for rooted trees. *Discrete Applied Mathematics*, 105 (1–3):147–158, 2000. doi: 10.1016/S0166-218X(00)00202-X.

Robert R. Sokal and Charles D. Michener. A statistical method for evaluating systematic relationships. *University of Kansas Science Bulletin*, 38(22):1409–1438, 1958. URL https://archive.org/details/universityofkans3802univ. No DOI.

Alexander Strehl and Joydeep Ghosh. Cluster ensembles – a knowledge reuse framework for combining multiple partitions. *Journal of Machine Learning Research*, 3:583–617, 2002. URL https://www.jmlr.org/papers/v3/strehl02a.html. No DOI.

Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, Evgeni Burovski, Pearu Peterson, Warren Weckesser, Jonathan Bright, Stéfan J. van der Walt, Matthew Brett, et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17(3):261–272, 2020. doi: 10.1038/s41592-019-0686-2. SciPy 1.0 Contributors.

D. Vu, M. Groenewald, M. de Vries, T. Gehrmann, B. Stielow, U. Eberhardt, A. Al-Hatmi, J. Z. Groenewald, G. Cardinali, J. Houbraken, T. Boekhout, P. W. Crous, V. Robert, and G. J. M. Verkley. Large-scale generation and analysis of filamentous fungal DNA barcodes boosts coverage for kingdom fungi and reveals thresholds for fungal species and higher taxon delimitation. *Studies in Mycology*, 92:135–154, 2019. doi: 10.1016/j.simyco.2018.05.001.

Josh Walker. pyitsx: fast ITS region detection and extraction using in-process HMM search. https://github.com/joshuaowalker/pyitsx, 2026. Python package, version 0.1.0.

Jiajie Zhang, Paschalia Kapli, Pavlos Pavlidis, and Alexandros Stamatakis. A general species delimitation method with applications to phylogenetic placements. *Bioinformatics*, 29(22):2869–2876, 2013. doi: 10.1093/bioinformatics/btt499.